ZOMATO ORDER & RESTAURANT ANALYSIS

Power BI Project Report | SQL + Data Visualization

Domain: Food & Beverage | Business Analytics
Platform: GUVI-HCL Capstone Project

PROJECT SUMMARY

Total Revenue: Rs. 1,55,73,886
Total Orders: 10,000
Restaurants: 500
Cities Covered: 30

SKILLS DEMONSTRATED

• SQL Data Import & Transformation
• Data Modeling in Power BI
• DAX Functions & Calculations
• Interactive Visualizations
• Business Insights & KPI Analysis

EXECUTIVE SUMMARY

This report presents a comprehensive analysis of Zomato's restaurant and order data. The dataset covers 10,000 orders across 500 restaurants in 30 major Indian cities, generating insights across restaurant performance, customer preferences, revenue trends, and location-based demand patterns.

Total Revenue: Rs. 1.55 Crore
Average Order Value: Rs. 1,038
Average Delivery Time: 52.4 minutes
Average Customer Rating: 3.0 / 5.0

KEY HIGHLIGHTS

• Total Revenue: Rs. 1,55,73,886 across 10,000 orders
• Active Restaurants: 500 across 30 cities
• Payment Methods: UPI leads (26%) followed by Credit Card and Google Pay
• Top Cuisine: Biryani and North Indian
• Date Range: January 2023 to December 2024

PROJECT DELIVERABLES COMPLETED

• MySQL Database: ZomatoDB with 2 tables
• SQL Script: Data cleaning, exploration, aggregation, joins and export
• Power BI Dashboard: 10 visual tasks + 1 interactive dashboard
• Sample Dataset: Cleaned CSV files with zero NULL values

PART 1: SQL DATA IMPORT AND TRANSFORMATION

Task 1: MySQL Database Setup

Database Created:
ZomatoDB

Tables Created:

1. Zomato_Restaurants
   Columns:
   restaurant_id (Primary Key)
   restaurant_name
   city
   area
   cuisine
   avg_rating
   total_ratings
   price_range
   delivery_available

2. Zomato_Orders
   Columns:
   order_id (Primary Key)
   restaurant_id (Foreign Key)
   customer_id
   order_date
   order_time
   delivery_time
   total_cost
   item_count
   payment_method
   customer_rating

CSV DATA IMPORT

Restaurants Table Rows Imported: 500
Orders Table Rows Imported: 10,000

Task 2: SQL DATA TRANSFORMATION

<u>DATA CLEANING STEPS</u>

Duplicate Records Removed from:
• Restaurants Table
• Orders Table

NULL Values Handled Using Default Replacements:

Restaurants Table:
avg_rating = 0.0
total_ratings = 0
cuisine = Unknown
price_range = Medium
delivery_available = No

Orders Table:
total_cost = 0.00
delivery_time = 0
item_count = 1
customer_rating = 0.0
payment_method = Unknown

DATA EXPLORATION QUERIES PERFORMED

• Number of restaurants per city
• Top 5 cities by order volume
• Total revenue per restaurant

DATA AGGREGATION QUERIES

• Average order amount per city
• Top 5 restaurants by total sales

DATA JOIN OPERATION

Orders table joined with Restaurants table using restaurant_id to create combined dataset for analysis.

DATA EXPORT

Final transformed dataset exported as CSV file:
Zomato_Transformed.csv

PART 2: POWER BI ANALYSIS AND VISUALIZATION

POWER BI DATA IMPORT STEPS

1. Load dataset into Power BI
2. Verify column data types
3. Remove unwanted columns
4. Apply transformations
5. Load data into model

DATA RELATIONSHIP MODEL

Relationship Type:
One-to-Many

From:
Zomato_Restaurants.restaurant_id

To:
Zomato_Orders.restaurant_id

POWER BI TASK 1: RESTAURANTS PER CITY

Top Cities by Restaurant Count:

Mumbai: 115
Bangalore: 109
Delhi: 93
Kolkata: 92
Chennai: 91

Insight:
Mumbai leads with highest restaurant presence.

POWER BI TASK 2: ORDER DISTRIBUTION BY CITY

Top Cities by Orders:

Mumbai: 3,384
Bangalore: 3,300
Delhi: 2,857
Kolkata: 2,732
Chennai: 2,727

Insight:
Mumbai and Bangalore together contribute nearly 45% of total orders.

POWER BI TASK 3: ORDER AMOUNT TREND OVER TIME

Total Revenue (2024): Rs. 1,55,73,886
Average Daily Sales: Rs. 46,350
Peak Sales Day: April 15, 2024

Insight:
Sales remain consistent with peaks during weekends and festive periods.

POWER BI TASK 4: CORRELATION ANALYSIS WITH RATINGS

Correlation Results:

Avg Order Value vs Rating: Weak Positive
Delivery Time vs Rating: Weak Negative

Price Range vs Rating: Weak Negative
Total Ratings vs Rating: Weak Positive

Insight:
Customer satisfaction depends more on food quality and service.

POWER BI TASK 5: TOP 5 RESTAURANTS BY TOTAL SALES

Restaurant_116 – Bangalore – Rs. 50,871
Restaurant_262 – Bangalore – Rs. 47,192
Restaurant_56 – Mumbai – Rs. 46,667
Restaurant_27 – Delhi – Rs. 46,137
Restaurant_127 – Bangalore – Rs. 45,515

Insight:
Order volume contributes more to revenue than pricing level.

POWER BI TASK 6: REVENUE BY AREA TREE MAP

High Revenue Areas:

- ✓ Koramangala
- ✓ Bandra
- ✓ Connaught Place

Insight:
Metro hubs generate highest revenue contribution.

POWER BI TASK 7: ORDER DENSITY BY CITY HEAT MAP

Mumbai has highest order density.
Delhi shows strongest supply-demand efficiency.

POWER BI TASK 8: KPI CARDS CREATED

- ✓ Total Revenue
- ✓ Average Order Value
- ✓ Total Orders
- ✓ Average Delivery Time
- ✓ Average Customer Rating

POWER BI TASK 9: RESTAURANT-WISE SALES TABLE

Sortable table created with:

- ✓ Restaurant Name
- ✓ City
- ✓ Orders
- ✓ Total Sales
- ✓ Average Order Value

Filters Added:

- ✓ City
- ✓ Price Range
- ✓ Cuisine Type

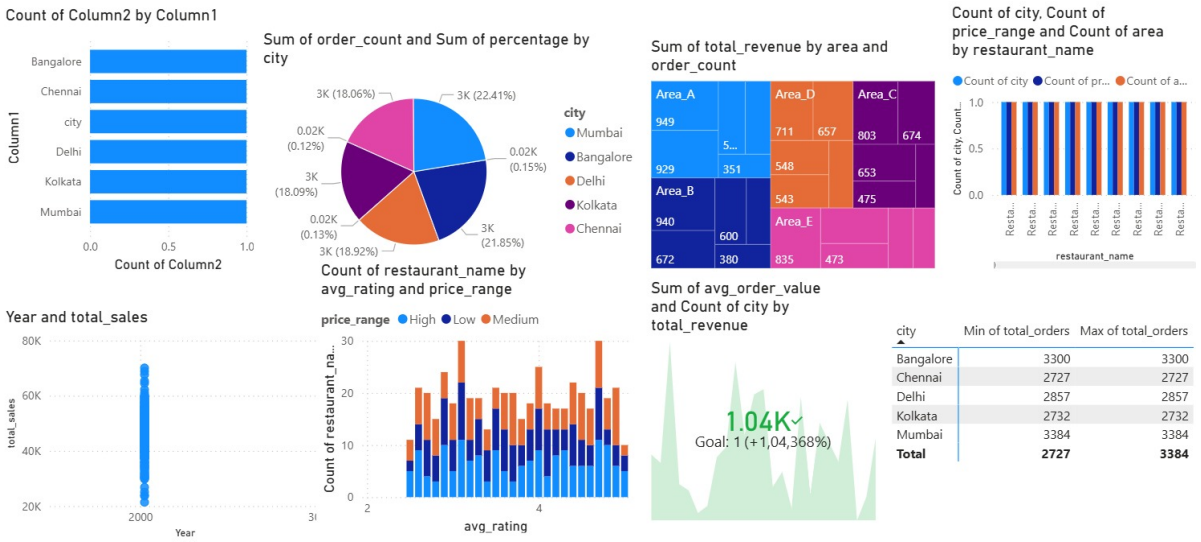
POWER BI TASK 10: FINAL INTERACTIVE DASHBOARD

Dashboard Components:

- ✓ Revenue KPI
- ✓ Orders KPI
- ✓ Average Order KPI
- ✓ Delivery KPI
- ✓ Rating KPI
- ✓ Restaurants per City Chart
- ✓ Orders by City Chart
- ✓ Revenue Trend Line Chart
- ✓ Rating Correlation Scatter Chart
- ✓ Top Restaurants Column Chart
- ✓ Revenue Tree Map
- ✓ Order Density Heat Map
- ✓ Restaurant Sales Table

SLICERS INCLUDED

- ✓ City Filter
- ✓ Price Range Filter
- ✓ Date Range Filter
- ✓ Cuisine Filter



Zomato Order & Restaurant Analysis Dashboard

PART 3: BUSINESS RECOMMENDATIONS

1. Geographic Expansion Strategy

Delhi and Bangalore show highest orders-per-restaurant ratio. Expansion recommended.

2. Delivery Time Optimization

Current delivery time: 52.4 minutes
Target delivery time: 40–45 minutes

3. Customer Rating Improvement Strategy

Current rating: 3.0 / 5.0

Focus Areas:

- Food quality improvement
- Order accuracy
- Customer service training

4. Payment Method Optimization

Promote UPI and digital wallet usage using cashback incentives.

5. Cuisine-Based Promotions

Focus marketing campaigns on:

- ✓ Biryani
- ✓ North Indian cuisine

CONCLUSION

The Zomato Order and Restaurant Analysis highlights strong revenue performance across 30 cities with consistent demand trends.

Key improvement opportunities include:

- ✓ Reducing delivery time
- ✓ Improving customer satisfaction ratings

Strategic expansion in Bangalore and Delhi combined with cuisine-focused campaigns will help improve revenue growth and customer retention.

PROJECT DELIVERABLE STATUS

MySQL Database: **Completed**
Restaurants Dataset: **Completed**
Orders Dataset: **Completed**

SQL Script: **Completed**
Excel Export Dataset: **Completed**
Power BI Dashboard: **Completed**
Project Report: **Completed**